

PROJECT 4 — WasteWise (Monu, Aditya)

Smart Waste Management & Recycling Portal

1. Project overview

Project name: WasteWise

Category: Sustainability / Waste Management / Civic Technology

Target users: Households, students, local communities, and organizations interested in responsible waste disposal.

Problem statement:

People may not know how to separate waste, where to dispose of specific materials, or how to participate in local recycling activities.

Proposed solution:

WasteWise is a waste management information portal that organizes recycling guidance, collection information, environmental education, and community initiatives.

The project can demonstrate how a digital interface could help users find relevant waste management information.

2. Pages you need to create

No.	HTML file	Page name	Purpose
1	<code>index.html</code>	Landing Page	Introduce WasteWise
2	<code>dashboard.html</code>	Sustainability Dashboard	Display sample waste management statistics
3	<code>waste-guide.html</code>	Waste Sorting Guide	Explain how different materials should be sorted
4	<code>collection.html</code>	Collection Schedule	Display a sample collection calendar
5	<code>recycling-centers.html</code>	Recycling Center Directory	Show sample recycling center listings
6	<code>community.html</code>	Community & Events	Present environmental events and initiatives

Page 1: Landing Page — index.html

Sections

A. Navbar

- WasteWise logo.
- Home.
- Waste Guide.
- Collection Schedule.
- Recycling Centers.
- Community.
- Get Started.

B. Hero section

Heading:

"Small Actions. Cleaner Communities."

Supporting text:

"Learn how to sort waste, discover recycling resources, and explore ways to reduce waste in your community."

Buttons:

- Explore Waste Guide.
- View Collection Schedule.

C. Waste categories

Create visual cards for:

- Organic waste.
- Paper and cardboard.
- Plastic.
- Glass.
- Metal.
- Electronic waste.

D. How it works

1. Identify your waste.

2. Check the sorting guidance.
3. Find a suitable disposal or recycling option.

E. Community impact

Display illustrative statistics, clearly marked as sample data.

F. Community initiatives

Show environmental campaigns and cleanup events.

G. Footer

Include links to the waste guide, collection schedule, recycling directory, and community pages.

Page 2: Sustainability Dashboard — **dashboard.html**

Goal: Create a dashboard for a household or community waste management program.

Sections

A. Sidebar

- Overview.
- Waste Guide.
- Collection Schedule.
- Recycling Centers.
- Community.
- Environmental Impact.

B. Header

- Page title.
- Location selector appearance.
- Profile area.

C. Summary cards

Metric	Illustrative value
Waste Sorted	24 kg
Recyclables Collected	12 kg
Organic Waste	8 kg

D. Waste category chart

Create a CSS visualization showing the proportion of sample waste categories.

E. Collection schedule preview

Display the next few scheduled collection dates using fictional demonstration data.

F. Recycling tips

Create short informational cards.

G. Community activity

Display sample cleanup or recycling initiatives.

CSS components

- Dashboard sidebar.
 - Summary cards.
 - CSS bar charts.
 - Schedule cards.
 - Waste category badges.
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Page 3: Waste Sorting Guide — [waste-guide.html](#)

Goal: Help users understand how different waste categories should be handled.

Sections

1. Page header.
2. Waste category navigation.
3. Organic waste guide.
4. Paper and cardboard guide.
5. Plastic guide.
6. Glass guide.
7. Metal guide.
8. Electronic waste guide.
9. Common sorting mistakes.
10. Frequently asked questions.

Example category card

Paper and cardboard

Include:

- Examples of common materials.
- Preparation guidance.
- Contamination warnings.
- A reminder to check local recycling rules.

Make it clear that accepted materials vary by local waste collection provider.

CSS components

- Category cards.
 - Two-column information sections.
 - Icon containers.
 - Warning callouts.
 - FAQ sections using `<details>` and `<summary>`.
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Page 4: Collection Schedule — `collection.html`

Goal: Display a sample collection calendar.

Sections

A. Page heading

"Collection Schedule."

B. Location selection appearance

Show a location field and an illustrative selected location.

C. Collection type legend

- General waste.
- Recycling.
- Organic waste.
- Electronic waste collection event.

D. Weekly schedule

Day	Collection type	Sample time
Monday	General waste	8:00 AM
Wednesday	Recycling	8:00 AM

Friday	Organic waste	8:00 AM
Saturday	Community collection event	10:00 AM

Label this schedule as fictional demonstration content.

E. Preparation checklist

- Separate materials according to local rules.
- Keep recyclable materials clean where required.
- Check the relevant collection instructions.
- Follow local disposal requirements.

F. Calendar view

Use CSS Grid to create a calendar-style layout.

Page 5: Recycling Center Directory — `recycling-centers.html`

Goal: Present a directory interface for recycling facilities.

Sections

1. Page title.
2. Search field appearance.
3. Material category filters.
4. Recycling center cards.
5. Center details.
6. Accepted materials.
7. Operating hours.
8. Map preview placeholder.

Recycling center card

Include:

- Center name.
- Address.
- Accepted materials.
- Opening hours.
- Contact information.
- Directions button appearance.

Use clearly fictional centers and addresses for the prototype, or replace them with verified local information if you later make the website public.

CSS components

- Directory cards.
 - Two-column listing layout.
 - Filter chips.
 - Map placeholder.
 - Contact information panel.
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Page 6: Community & Events — **community.html**

Goal: Encourage community participation in environmental activities.

Sections

A. Hero banner

"Make a Difference Together."

B. Upcoming events

Display event cards with:

- Event title.
- Date.
- Location.
- Description.
- Participation button appearance.

C. Community projects

Examples:

- Neighborhood cleanup.
- School recycling campaign.
- Plastic reduction initiative.
- Community composting program.

D. Volunteer information

Explain how community participation might work.

E. Community highlights

Display sample achievements and project summaries.

F. Frequently asked questions

Explain the event participation process using static information.

Page 7: Environmental Impact — **impact.html**

Goal: Present illustrative environmental information through visualizations.

Sections

1. Page header.
2. Waste reduction summary cards.
3. Waste category breakdown.
4. Monthly collection chart.
5. Recycling progress visualization.
6. Community activity summary.
7. Environmental education section.
8. Data methodology and limitations.

Sample visualization

Display a fictional monthly waste reduction dataset.

Month	Waste recycled
January	120 kg
February	145 kg
March	170 kg
April	190 kg

Use CSS bars to visualize the values.

Explain that these are sample figures, not verified environmental impact measurements.

WasteWise design system

Element	Recommendation
Primary color	Forest green
Secondary color	Mint

Background	Off-white
Cards	White
Typography	Clean sans-serif
Border radius	12–16px
Visual style	Friendly, eco-conscious, community-oriented

Inspiration: Explore recycling and waste management interfaces to study their service cards, educational content, and collection information layouts.